Red Oasis | Sharifa's Field Research Dossier

NASA Space Apps Challenge 2026 — Week 2 deliverable
Prepared: 24 October 2026
Study crop: Date palm (Phoenix dactylifera)
Project coordinate supplied in the team roadmap: 23.308639 N, 56.613083 E
Status: The team point is in the Ad Dariz agricultural area of Ibri. Satellite imagery confirms cultivated plots around it, but the point is close to a plot/access boundary, so the exact orchard and its area remain unconfirmed. A separate 2007 mapping paper places Falaj al-Dariz near 23°19′N, 56°36′E; its 56°47′E coordinate is for Falaj al-Nujaid, not Dariz.

Executive summary

Red Oasis proposes a water-balance decision-support prototype for date-palm cultivation in the Ibri area of Al Dhahirah, Oman. A peer-reviewed Sultan Qaboos University study of Falaj al-Dariz reports an annual date-palm crop evapotranspiration (ETc) of **3,620 mm/year** and annual field supply including effective rainfall of **3,235 mm/year** for its 2002–2003 study cycle. The historical annual difference is **385 mm/year** (3,850 m3/ha/year), equivalent to **10.6% of ETc**; the study's demand/supply ratio is 1.12. Its monthly evidence indicates shortage from April through November and excess supply in December through February, with March in the accepted range. These are historical study results, not current measurements for the team's exact point.

The project point (23.308639, 56.613083) is about 1.45 km southwest of the Ad Dariz village point listed by OpenStreetMap-derived GeoView (23.32000, 56.62000). High-resolution Esri World Imagery, identified at the point as Vantor/WorldView-3 imagery dated 8 May 2024 (0.31 m source resolution; stated positional accuracy 5 m), shows irrigated-looking plots and dense tree cover around the point. The pin appears near a track/parcel edge rather than clearly inside one identifiable orchard. This verifies an agricultural oasis landscape in the vicinity, not a particular grower's field, crop species, or property boundary. The coordinates are consistent when the two aflaj are separated: a 2007 map labels al-Dariz at approximately 23°19′N, 56°36′E and al-Nujaid at 23°28′N, 56°47′E. The historic 255.44 ha area is the wider al-Dariz falaj cultivated/command area reported from a 2001 GPS/GIS survey, not the acreage of the selected point or a current satellite-measured boundary.

For the Mars concept, the 50%, 75%, and 100% cases below are transparent **design scenarios** against the Earth study's annual crop-water benchmark. They are not measured Martian agricultural values. NASA sources support closed-environment plant growth and water recovery as research directions, but do not establish these three percentages as Mars operating data.

1. Experimental site and evidence status

Field	Research finding	Evidence/status
Name in team roadmap	Ad Dariz / الدريز, Ibri, Al Dhahirah	Community-map data place the point in the village vicinity; use "Ad Dariz area, Ibri" pending orchard-level confirmation.
Coordinates for current project map	23.308639, 56.613083	Taken from the team roadmap. About 1.45 km southwest of the mapped Ad Dariz village node (23.32000, 56.62000); exact grove not confirmed.

Field	Research finding	Evidence/status
Map link / imagery check	Open point in Google Maps · Download marked satellite image	Esri World Imagery identifies the local image as Vantor/WorldView-3, 2024-05-08, 0.31 m source resolution. It shows cultivated plots around the point; the marker is near a track/parcel edge.
Coordinate cross-check	2007 map: al-Dariz ≈23°19′N, 56°36′E; al-Nujaid ≈23°28′N, 56°47′E; OSM-derived Ad Dariz village node: 23°19′12″N, 56°37′12″E	The earlier apparent longitude conflict came from assigning al-Nujaid's coordinate to al-Dariz. Use the team roadmap point for the prototype and keep the exact orchard parcel unconfirmed.
Historic falaj cultivated/command area	2,554,423 m² = 255.44 ha (2001 survey)	MRMEWR (2001) area as reported by Al-Ghafri; estimated by GIS after GPS land survey. This is the historical falaj-wide area, not a measurement of the current selected orchard.
Satellite boundary and area today	Not established	Imagery confirms cultivated land around the point, but no single orchard polygon can be assigned to the point. Acreage is intentionally left blank until the team selects and traces the intended parcel.
Main crop	Date palm (Phoenix dactylifera)	More than 90% of the cropping area in the study's Dariz and Nujaid systems was date palm. Applies to the historical study area.

Site-verification action before final submission

1. Open the map link and choose **Satellite**.
2. Confirm whether the pin lands on the intended palm-growing area. Compare the pin with visible green palm blocks and the named settlement/falaj.
3. If it does not land in the intended grove, right-click the centre of the target green area and copy the corrected latitude/longitude.
4. Trace only the continuous planted area as a Google Earth polygon. Save the polygon/KML and capture a screenshot showing the pin or polygon, place label, coordinates, and scale bar.
5. Keep the roadmap point provisionally, but label it "Ad Dariz area, Ibri" until the satellite check confirms the grove. Do not reuse 255.4 ha as today's site area.

2. Crop water demand and seasonal evidence

Historical annual water balance (Falaj al-Dariz study)

Quantity	Value	Meaning
Date-palm ETc	3,620 mm/year	Crop evapotranspiration demand reported from the Omani crop-water guide (Al-Nadi, 2003), for an assumed density of 1 palm per 15 m2.
Field supply, including effective rainfall	3,235 mm/year	Historical estimate for the 2002–2003 measurement period.
Demand minus supply	385 mm/year	Deficit in water-depth terms; 3,850 m3/ha/year.
Demand/supply ratio	1.12	Demand was 112% of supply; equivalently, the shortfall was about 10.6% of demand.
Mean rainfall context	Ibri: 88 mm/year; Tanam: 85 mm/year (1975–1999)	The paper reports no rain gauge at Dariz for this long-term record. Do not treat this as current rainfall at the selected point.

Monthly date-palm water demand

The monthly mean daily ETc values below are from the Al-Ghafri monthly series for May 2002–April 2003. Monthly depths are calculated as daily ETc multiplied by calendar days in that historical month; rounding causes the annual total to be 3,619.6 mm, consistent with the paper's rounded 3,620 mm/year. The per-hectare volume conversion is 1 mm = 10 m3/ha. These values are a historical reference baseline, not a present-day irrigation prescription.

Month in study cycle	ETc (mm/day)	Calculated ETc (mm/month)	Equivalent (m3/ha/month)	Observed D/S
May 2002	11.9	368.9	3,689	1.23
June 2002	14.7	441.0	4,410	1.52
July 2002	14.9	461.9	4,619	1.51
August 2002	13.8	427.8	4,278	1.45
September 2002	12.8	384.0	3,840	1.33
October 2002	11.2	347.2	3,472	1.22
November 2002	9.5	285.0	2,850	1.12
December 2002	4.0	124.0	1,240	0.49
January 2003	4.7	145.7	1,457	0.58
February 2003	4.4	123.2	1,232	0.56
March 2003	6.9	213.9	2,139	0.90
April 2003	9.9	297.0	2,970	1.30
Annual total (rounded)	—	**3,619.6**	**36,196**	**1.12 annual**

D/S interpretation: The study defines demand/supply greater than 1.0 as insufficient supply, below 1.0 as supply exceeding crop demand, and 0.6–1.0 as an accepted range for surface irrigation. Thus its 2002–2003 result indicates a seasonal mismatch: under-supply April–November, over-supply December–February, and an in-range March. The crop ETc table and the irrigation-supply series are historical and specific to that study cycle.

Date-fruit growth stages

The FAO sequence below describes fruit development after fertilisation. Stage timing varies with cultivar, pollination date, and local conditions; no cultivar-specific calendar for the selected grove has yet been confirmed.

Stage	Approximate duration / description
Hababouk	Begins after fertilisation; about 4–5 weeks; very young fruit.
Kimri	Green, hard fruit; longest stage, about 9–14 weeks depending on variety.
Khalal	Colour change and physiological maturity; about 3–5 weeks.
Rutab	Softening and browning/ripening; about 2–4 weeks.
Tamar	Fully ripe/final stage; fruit on a bunch can ripen over nearly a month, requiring multiple harvests.

3. Soil-moisture thresholds: what can and cannot be claimed

FAO date-palm irrigation guidance describes scheduling in terms of **depletion of total available soil water (TAW)** and gives 40% depletion as a general scheduling fraction in its worked example. That is **not** a universal volumetric soil-moisture percentage. Al-Ghafri reports Dariz soil dominated by clay loam, bulk density 1.43 g/cm3, and field capacity 22% by volume; field capacity alone is not enough to derive TAW because the permanent wilting point and calibrated root-zone profile are also needed.

The app no longer uses 30 / 45. Its current `soil_dry = 8.4` and `soil_moderate = 9.3` are the 25th and 75th percentiles of the Ad Dariz SMAP record, and the provisional rule compares with the site median (about 8.8%). They describe dry or normal for this site's satellite record and must remain labelled **illustrative demo values**, not locally validated thresholds. A defensible next step is to calibrate soil moisture in the active root zone, determine field capacity and wilting point for the exact plot, then define thresholds on a clearly stated scale (for example, % TAW remaining).

The fixed `temp_high = 40°C` and `ndvi_low = 0.30` are also not validated by the Dariz study as irrigation triggers. Keep them as demo rules or replace them only after site/season calibration. NDVI should preferably be interpreted relative to the same grove's seasonal baseline and sensor/product, not by borrowing an unvalidated universal cutoff.

4. Mars closed-environment concept and scenario calculations

NASA research describes controlled plant-growth chambers, LED lighting, recirculating hydroponics, humidity recovery, and water recovery for life-support systems. These sources establish technical concepts and research direction; they do not supply a verified date-palm irrigation demand for Mars. The Mars tab should therefore be described as a **conceptual resource-budget scenario**.

Assumption and equation

- Reference crop demand: 3,620 mm/year, borrowed from the Earth Falaj al-Dariz study solely as a normalisation baseline.
- Scenario supply fraction: 50%, 75%, or 100% of that reference demand.
- Water balance = scenario supply minus crop demand. Negative values indicate an unmet amount in this simplified model.
- Recycling is described qualitatively. The scenario percentages are not NASA water-recovery performance values and do not represent Martian field measurements.

Scenario	External/available water budget	Supply (mm/year)	Supply (m3/ha/year)	Demand (mm/year)	Simplified balance (mm/year)	Unmet demand
Restricted	50% of reference	1,810	18,100	3,620	-1,810	50%
Limited	75% of reference	2,715	27,150	3,620	-905	25%
Full reference	100% of reference	3,620	36,200	3,620	0	0%

Interpretation: In a real closed habitat, irrigation water recirculation, transpiration capture, leaks, crop choice, rooting media, harvest area, power, nutrients, and life-support needs determine make-up water. These simple values only let judges compare constrained versus unconstrained water budgets. They should not be presented as an engineering design or actual Mars crop-water requirement.

5. Site climate summary and evidence limits

NASA POWER's official climatology endpoint returns a 25-year meteorological and solar climatology for January 2001–December 2025. For the roadmap point, mean 2 m air temperature is 27.75°C annually; the monthly mean peaks at 35.76°C in July and is lowest at 17.58°C in January. Mean daily all-sky surface shortwave radiation is 22.03 MJ/m²/day annually, with monthly means shown below. These are gridded model/satellite-derived climate values, not orchard sensor readings. POWER reports the selected grid elevation as 607.04 m and the underlying meteorological grid as 0.5° × 0.625°; it is appropriate for regional context, not field-scale soil moisture or microclimate.

Month	NASA POWER T2M climatology (°C)	All-sky shortwave radiation (MJ/m²/day)
January	17.58	16.63
February	19.85	19.47
March	23.73	22.57
April	28.63	24.91
May	33.11	27.15
June	35.30	26.95
July	35.76	24.65
August	34.88	24.20
September	32.58	23.25
October	28.59	21.11
November	23.26	17.61
December	19.27	15.80

Recent local rainfall/temperature comparison: The Oman National Centre for Statistics and Information weather portal lists Ibri station monthly observations for 2025. The station's 2025 monthly mean temperature ranges from 19.3°C (January) to 38.4°C (July); monthly rainfall sums to 176.4 mm, with 54.8 mm in March, 35.0 mm in April, and 59.0 mm in October. This station is a useful local comparison, not a gauge at the selected oasis point. 2025 monthly source data

Al-Ghafri reports historical mean annual rainfall of 88 mm at Ibri and 85 mm at Tanam during 1975–1999, and effective rainfall of 44.1 mm in the 2002–2003 study accounting period. These records use different periods and sources, so they should not be merged into a single "normal." No recent measured irrigation volume specific to the selected Dariz parcel has been found. The latest detailed Oman date-palm irrigation comparison located is a 2022 publication reporting experiments conducted in 2015–2016 at Al Kamil and Al Wafi in South Al Sharqiyah; treat it as a modern-method comparison, not Dariz field data.

6. Published modern irrigation comparison (Oman; not Dariz)

Dhehibi et al. (2022) analyze Oman station experiments measured in 2015–2016 with Khalas palms. At 100% crop water requirement, bubbler irrigation used **9,966.154 m³/ha/year** and yielded **79.0 kg/palm**. Subsurface drip at 60% ETc used **6,420.600 m³/ha/year** and yielded **78.3 kg/palm**; the paper reports no significant yield difference between these two treatments. That is **3,545.554 m³/ha/year (35.6%) less water** in that experiment. At 40% and 20% ETc, yields fell to 69.8 and 68.8 kg/palm, so those severe reductions should not be presented as yield-neutral. The irrigation system's emitters were buried at 40 cm and water applied daily based on nearby weather data. The trial was at Al Kamil and Al Wafi Agricultural Research Station in South Al Sharqiyah, not Dariz, and the measurements are from 2015–2016 despite the 2022 publication.

Oman trial treatment	Water (m³/ha/year)	Yield (kg/palm)	Interpretation
Bubbler, 100% water requirement	9,966.154	79.0	Trial baseline
Subsurface drip, 60% ETc	6,420.600	78.3	35.6% less water; reported yield not significantly different
Subsurface drip, 40% ETc	4,239.704	69.8	Lower water; yield reduced
Subsurface drip, 20% ETc	2,400.681	68.8	Lower water; yield reduced

Downloadable data table: Oman date-palm irrigation comparison CSV.

Project implication: cite 60% ETc subsurface drip as a tested Oman comparator and a candidate for future evaluation, not as a prescription for Dariz. Actual scheduling needs crop stage, soil, salinity, water source and field measurements.

7. What the project can claim and what it cannot

Supported claims: Date palm was the dominant crop in the historical Dariz falaj study; the paper reports 3,620 mm/year ETc and 3,235 mm/year supply including effective rainfall for that study; historical seasonal demand and supply did not match; NASA has researched controlled plant-growth systems and water recovery for spaceflight.

Limits: Satellite imagery does not identify the exact crop or orchard boundary at the project pin. The 255.44 ha figure is the historical falaj-wide 2001 survey, not the selected orchard's current area. Dariz irrigation measurements are historical (2002–2003); current parcel-level irrigation values were not found in published sources. The app's demo thresholds are not locally calibrated. Mars percentages are scenarios, not measured Mars conditions. No farmer interview was conducted, and no farmer-sourced figures are claimed; interviews are optional.

Recommended status wording: "Satellite-reviewed candidate point in the Ad Dariz agricultural area, Ibri; exact orchard polygon/area, grove crop identity at the pin, irrigation practice, and sensor thresholds remain to be confirmed locally. Regional climate context is from NASA POWER and Ibri station observations."

8. References

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2. Al-Ghafri, A. (2006). "Aflaj Irrigation D/S Ratio" (conference/manuscript source containing the monthly date-palm ETc and D/S series used in the table above). ResearchGate record. The peer-reviewed SQU article is the primary citation for the annual balance and published study findings.

3. Liebenberg, P.J. & Zaid, A. “Chapter VII: Date Palm Irrigation.” FAO, Date Palm Cultivation: FAO chapter
4. Zaid, A. & de Wet, P.F. “Chapter I: Botanical and Systematic Description of the Date Palm.” FAO, Date Palm Cultivation: FAO chapter
5. NASA. “Space Crops.” NASA Science: NASA page
6. NASA. “Advanced Plant Habitat.” NASA Science: NASA page
7. NASA. “Environmental Control and Life Support Systems (ECLSS).” NASA page
8. NASA TechPort. “MarsOasis - An Efficient Autonomously Controlled Martian Crop Production System.” Project page
9. OpenStreetMap-derived GeoView. “Ad Dariz ,(الدريز) Ibri,” mapped at 23°19’12”N, 56°37’12”E; OSM node 1721807540 (map data last updated 2018-10-28). GeoView record | OSM node. Used only to locate the village vicinity; not proof of current orchard limits.
10. NASA POWER. “Climatology API” and “Temporal APIs.” Climatology and monthly averaging documentation. Point request: 23.308639 N, 56.613083 E; 25-year period Jan 2001–Dec 2025; T2M and ALLSKY_SFC_SW_DWN. Raw climatology CSV | 2001–2025 monthly series CSV | clean monthly summary.
11. Oman National Centre for Statistics and Information. “Weather — Adh Dhahirah, Ibri station,” 2025 monthly mean temperature and rainfall. Rainfall series | Mean-temperature series. Compiled monthly values: CSV.
12. Al-Ghafri, A. (2007). “Equitability of Water Distribution in Aflaj Irrigation Systems of Oman.” Map identifies al-Dariz near N23°19′, E56°36′ and al-Nujaid near N23°28′, E56°47′. Conference paper. Area attributed to MRMEWR (2001), GIS after GPS land survey.
13. Dhehibi, B. et al. (2022). “Economic and environmental evaluation of different irrigation systems for date palm production in the GCC countries: the case of Oman and Saudi Arabia.” Oman measurements were 2015–2016, Al Kamil and Al Wafi; Khalas date palm. Open report PDF | DOI.
14. Esri, Vantor, Earthstar Geographics and GIS User Community. World Imagery basemap description. Point-specific ArcGIS identify response returned Vantor/WorldView-3, capture date 2024-05-08, source resolution 0.31 m and stated positional accuracy 5 m. Marked image.

Prepared for the Red Oasis `docs/` folder. The marked satellite image, NASA POWER 25-year climatology, monthly NASA POWER time series, official 2025 Ibri station comparison, and Oman irrigation-comparison CSV are included alongside this dossier. Published evidence is labeled by site and year. No interview is required for this submission.